

2 BIT COMPARATOR

2-Bit Magnitude Comparator — Inputs & Outputs				
Type	Signal	MSB	LSB	Width
Input	A[1:0]	A[1]	A[0]	2-bit
Input	B[1:0]	B[1]	B[0]	2-bit
Output	A_greater_B	—	—	1-bit
Output	A_equal_B	—	—	1-bit
Output	A_less_B	—	—	1-bit

MAIN CODE

```
module magnitude_comparator_2bit (  
    input wire [1:0] A,  
    input wire [1:0] B,  
    output reg A_greater_B,  
    output reg A_equal_B,  
    output reg A_less_B  
);  
  
always @(*) begin  
  
    if (A > B) begin  
        A_greater_B = 1'b1;  
        A_equal_B  = 1'b0;  
        A_less_B   = 1'b0;  
    end  
  
    else if (A == B) begin
```

```

        A_greater_B = 1'b0;
        A_equal_B  = 1'b1;
        A_less_B   = 1'b0;
    end

    else begin
        A_greater_B = 1'b0;
        A_equal_B  = 1'b0;
        A_less_B   = 1'b1;
    end

end

endmodule

TEST BENCH CODE

module magnitude_comparator_2bit_tb;

    reg [1:0] A;
    reg [1:0] B;

    wire A_greater_B;
    wire A_equal_B;
    wire A_less_B;

    // Instantiate the Magnitude Comparator
    magnitude_comparator_2bit uut (

```

```
.A(A),  
.B(B),  
.A_greater_B(A_greater_B),  
.A_equal_B(A_equal_B),  
.A_less_B(A_less_B)  
);
```

```
initial begin
```

```
// A = 00, B = 00
```

```
A = 2'b00;
```

```
B = 2'b00;
```

```
#10;
```

```
// A = 00, B = 01
```

```
A = 2'b00;
```

```
B = 2'b01;
```

```
#10;
```

```
// A = 00, B = 10
```

```
A = 2'b00;
```

```
B = 2'b10;
```

```
#10;
```

```
// A = 00, B = 11
```

```
A = 2'b00;
```

```
B = 2'b11;
```

```
#10;
```

```
// A = 01, B = 00
```

```
A = 2'b01;
```

```
B = 2'b00;
```

```
#10;
```

```
// A = 01, B = 01
```

```
A = 2'b01;
```

```
B = 2'b01;
```

```
#10;
```

```
// A = 01, B = 10
```

```
A = 2'b01;
```

```
B = 2'b10;
```

```
#10;
```

```
// A = 01, B = 11
```

```
A = 2'b01;
```

```
B = 2'b11;
```

```
#10;
```

```
// A = 10, B = 00
```

```
A = 2'b10;
```

```
B = 2'b00;
```

```
#10;
```

```
// A = 10, B = 01
```

```
A = 2'b10;
```

```
B = 2'b01;
```

```
#10;
```

```
// A = 10, B = 10
```

```
A = 2'b10;
```

```
B = 2'b10;
```

```
#10;
```

```
// A = 10, B = 11
```

```
A = 2'b10;
```

```
B = 2'b11;
```

```
#10;
```

```
// A = 11, B = 00
```

```
A = 2'b11;
```

```
B = 2'b00;
```

```
#10;
```

```
// A = 11, B = 01
```

```
A = 2'b11;
```

```
B = 2'b01;
```

```
#10;
```

```
// A = 11, B = 10
```

```
A = 2'b11;
```

```
B = 2'b10;
```

```
#10;
```

```
// A = 11, B = 11
```

```
A = 2'b11;
```

```
B = 2'b11;
```

```
#10;
```

```
$finish;
```

```
end
```

endmodule